

Jose Bande Serrano

Full-stack Software Engineer

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Professional Summary

Full-stack Software Engineer with 17+ years of experience building AI-powered applications, workflow automation, and enterprise business systems. Hands-on background designing AI agents, multi-step automation workflows, API-integrated services, and governed compliance solutions across Azure, OpenAI, Microsoft Teams, Power Automate, React, Next.js, and Node.js. Brings end-to-end ownership from solution design and stakeholder alignment through deployment, measurement, and continuous improvement in enterprise environments.

Selected Skills

AI Agents, OpenAI, Workflow Automation, REST APIs, Node.js, Azure Functions, Power Automate, Dataverse, Microsoft Teams, React, Next.js, TypeScript, Azure, CI/CD, System Architecture, Project Management

Professional Experience

NBCUniversal - Full-Stack Engineer, Compliance Automation

2024-09 - present | New York, NY

Built and delivered a production compliance automation platform combining a high-performance Next.js SPA, backend workflows, ETLs, and reporting systems.

- Built a complex User Access Review compliance system on the Power Platform for NBCUniversal.
- Created dozens of Power Automate flows and dataflows to automate compliance workflows and data movement.
- Designed and implemented tens of Dataverse tables to support the application's governed enterprise data model.
- Implemented AI bots integrated with Microsoft Teams and supported engineering workflow automation and delivery.
- Built a React.js and Next.js application that is in production and used by hundreds of NBCUniversal employees.

Enerex - AI Development Consultor

2023-11 - present | Remote - New Jersey

Architected AI-powered automation platforms spanning user-facing applications, asynchronous workflow orchestration, and cloud delivery for contract-processing systems.

- Architected the "Unattended Mode" platform, an asynchronous work-farm architecture for AI-driven email and contract processing without requiring users to wait in front of a UI.
- Designed a multi-step automation solution around a connector API, monitoring application, and queue-triggered Azure Functions workers to intake tasks, orchestrate flows, and deliver results by webhook.
- Built an AI-powered Outlook add-in to automate contract processing workflows for energy retail market users and support user review before downstream submission.
- Implemented a queue-driven execution model using Azure Storage Queues and Azure Functions so tasks could be processed in parallel and the platform could scale horizontally under background workload demand.
- Provisioned and deployed Azure Functions, web applications, databases, and Storage Queues to support Enerex products.

Broadcasting Platforms - Senior Frontend Engineer & Team Lead

2019-11 - 2022-12 | Remote

Led and built the core user-facing frontend applications of a real-time streaming platform, with deep ownership across Vue.js architecture, design-system implementation, and WebSocket-driven interaction systems.

- Improved frontend delivery performance by introducing HTTP/2, asset compression, and image conversion to WebP format.
- Built two core user-facing streaming applications in Vue.js that formed the frontend backbone of the platform.
- Designed and implemented an SEO-focused rendering solution that served server-side-rendered content to web crawlers while preserving the SPA experience for regular users.
- Implemented demanding frontend experiences from professional Figma design specifications with strong attention to design fidelity and usability.
- Used Storybook as a design-system layer to centralize, standardize, and control component styling across the platform.

First Due - Senior Software Engineer

2017-07 - 2019-11 | Remote

Built large-scale web crawlers and ETL pipelines with Python and event-driven backend systems.

- Designed and maintained PostgreSQL databases supporting large-scale data collection workloads.
- Implemented tens of Python web crawlers with the Scrapy framework to extract critical company data from public websites.
- Architected a system to manage and control web crawlers, centralizing execution control and operational monitoring in a single UI.
- Built crawler orchestration and monitoring capabilities using Django, FastAPI, and RabbitMQ.
- Improved control and accountability of company web-crawling operations through centralized management and visibility.

Projects

AI Assisted Invoice Processor

An application to help finance teams collect invoices, extract structured data, review exceptions, and move approved records into downstream reporting without juggling disconnected inboxes, sheets, and shared folders.

- Implemented AI-powered invoice analysis using OpenAI.
- Built the backend with Node.js, Express.js, and Prisma over SQL Server hosted in Azure.
- Integrated email-based ingestion flows so incoming documents could be collected and processed automatically.
- Built the frontend with React.js, Next.js, Tailwind CSS, Lucide React, and TanStack to support real-time, data-driven invoice operations.
- Deployed the system in Azure with CI/CD through Azure DevOps.

Education

Bachelor's Degree, Engineer in Electronics

Ciudad Universitaria Jose A. Echeverria (CUJAE) | Havana, Cuba | 2004 - 2009